Resistance Training and Muscular Load: Your Muscles Don't Count Minutes

A strength workout is a bit awkward to score because the easiest information to collect is also some of the least useful. "Strength training for 45 minutes" could describe a circuit with almost no rest, 5 heavy sets of squats, a long technical session with plenty of standing around, or a few light isolation exercises. Those sessions may have the same duration and a similar average heart rate while leaving the user with very different cardiovascular and muscular fatigue.

The first useful distinction is between internal and external load. Internal load is the body's response to the session: heart rate, oxygen demand and perceived effort. External load is the work that was performed: the exercise, load, repetitions, distance, speed, gradient or power. They are related, although one can't reliably stand in for the other across every kind of exercise. Research comparing common training-load methods has found strong relationships between external work and oxygen consumption during controlled endurance sessions, while heart-rate and perceived-effort methods remain measures of the individual response to that work. [1] This gives us 2 useful accounts of a workout rather than one perfect number.

Cardiovascular load is the clearer half. Google's Cardio Load model follows the TRIMP idea and accumulates load from heart-rate reserve (HRR), meaning the user's heart rate relative to the space between resting and maximum heart rate. Each minute becomes progressively more valuable as HRR rises, so 10 hard minutes count much more than 10 easy minutes. [2] This remains a sensible way to calculate the cardiovascular part of lifting. A dense leg circuit should accumulate more cardio load than a long-rest bench session if it genuinely keeps the cardiovascular system working harder.

Heart rate still needs some care during resistance exercise. Lifting is intermittent, the hardest muscular effort may last only a few seconds, and the cardiovascular response includes large changes in blood pressure which a wrist wearable does not measure. Breath holding and the Valsalva manoeuvre (forcefully exhaling against a closed airway, common during heavy lifting) add another response that is unlike steady aerobic exercise. [3] Wrist optical heart-rate accuracy also becomes less reliable during rapid and sprint-like movement for some devices. [4] Heart rate can estimate aerobic energy cost reasonably well in controlled low-intensity resistance exercises, but that evidence only came from healthy trained men performing continuous 4-minute bouts at only 12-24% of 1RM (1 rep max). [5] It should not be extrapolated into the claim that HR captures the full cost of a heavy strength session.

Exercise logs help cardiovascular load mainly by improving context. They tell us when a workout happened, whether apparently quiet periods were rest between sets, and whether a poor accelerometer signal should be allowed to erase a real exercise bout. They should not create imaginary heart beats from the number of kilos lifted. When reliable heart-rate data exists, cardio load should still come from the HRR curve. Logged exercise, distance and power become validating external evidence and a fallback when the heart-rate channel is missing or clearly poor.

The muscular side is more difficult because "muscular load" is not one settled physiological quantity. It can mean mechanical work, local fatigue, muscle damage, hypertrophy stimulus (the amount of stimulus an exercise provides for muscle growth), loss of force or the recovery cost that follows the session. Those outcomes overlap but do not move together. A heavy set may create a large strength-specific stimulus with few repetitions. A light set taken close to failure may create substantial local and metabolic fatigue. Volume load, usually sets × repetitions × external weight, is useful for describing what was lifted, but it says about the muscles involved, range of motion, relative intensity or how hard the set was for that person. [6]

The literature makes this limitation fairly obvious. When volume load is held equal, heavier training tends to produce greater gains in 1RM strength, while hypertrophy can be similar across a wide range of loads. [7] Counting hard sets can be a reasonable volume measure for hypertrophy when sets are taken to or near failure and the remaining conditions are similar. [8] The words "near failure" are doing a lot of work there. Training to failure creates more acute performance loss, metabolic response, muscle damage and perceived effort than stopping earlier, even where the lifted work can look similar. [9] A recent review also found set duration, proximity to failure, total volume and training density to be major determinants of acute fatigue. [10]

This means that exercise, weight and reps give a strong external-load estimate, but they still leave one important thing unknown: how many repetitions the user could have completed. A single post-session effort question would therefore add useful information. RPE (Rating of Perceived Exertion, a scale where you rate how hard the exercise felt) has shown a strong overall relationship with resistance-exercise intensity and physiological exertion, although there is large variation between studies and users need some familiarity with the scale. [11] Session RPE also responds to work rate; completing the same work with shorter rests would generally feel harder. [12] If effort is not collected, the model should calculate from the available work and report lower confidence rather than just assuming every set reached failure.

Relative intensity gives the exercise log a personal scale. For each exercise, the entered load can be divided by a recent estimate of 1 rep max (e1RM). Multiple-repetition tests can estimate 1RM reasonably well, especially from sets of 10 or fewer reps, although the estimate is exercise-specific and assumes the set was reasonably close to the user's maximum capacity. [13] In normal logging that assumption will often be false. The safest implementation is to treat a set-derived e1RM as a lower bound, use the best credible estimate from recent top sets, and let it fall slowly rather than interpreting one light day as lost strength. A direct 1RM, where genuinely available, is more reliable and 1RM testing itself generally shows good test-retest reliability. [14]

The raw set calculation can then be kept fairly simple:

$$\text{Set dose} = \text{repetitions} \times \text{relative load} \times \text{exercise involvement} \times \text{set difficulty}$$

Relative load is effective load divided by the user's current exercise specific e1RM. Exercise involvement comes from an exercise library rather than the workout label. A small isolation movement starts below a large upper-body compound movement, which starts below a large lower-body or full-body movement. This is not because a squat repetition is magically worth a fixed number of curl repetitions. It is a practical way to acknowledge the amount of muscle involved until the coefficients can be calibrated from real sessions. Sensible starting bands would be around 0.6 for small isolation exercises, 0.8 for larger single-joint movements, 1.0 for multi-joint upper-body movements and 1.2 for lower-body or full-body compound movements.

Set difficulty should be 1.0 when no effort information exists. If session RPE is collected, a restrained modifier can move the final muscular dose up or down, initially no more than about 20%. If repetitions in reserve (RIR) is added in future, it can operate at set level and carry much more information: sets ending around 0-3 RIR receive the full difficulty factor, while clearly easy sets are reduced. The exact curve should be validated because people are imperfect at predicting repetitions to failure, particularly when they stop far from it. [15]

The exercise library should also allocate each set across muscle groups. A squat may assign most of its dose to the quads and glutes with smaller shares elsewhere; a bench press may assign its dose across the chest, triceps and anterior delts. Primary and secondary shares are enough for the first version. This prevents 6 sets of squats and 6 sets of curls from looking like the same muscular exposure, and it allows the system to show that 2 sessions loaded different tissues even where their total points were similar. The session total should sum these local doses with a modest diminishing return transform, such as the square root or logarithm after a minimum useful dose. Otherwise extremely high repetition sessions can make the score grow without limit even while the added information from every extra repetition becomes less convincing.

Loads also need standardisation before this equation is used. Dumbbell weight must state whether it is per hand or total. Unilateral repetitions should be recorded per side. Machines need separate exercise identities because 80 kg on 2 machines is not the same resistance. Assisted exercises should subtract assistance rather than record it as lifted load. Drop sets should remain separate sets because their relative intensity changes. Warm-up sets can be included with their real lower contribution, or explicitly marked as warm-ups and heavily reduced.

Bodyweight exercises can't be assigned zero load. Effective load should be estimated from body mass and an exercise specific biomechanical coefficient, with different variants treated as different exercises. Push-ups and bench press can produce similar upper-body activation at comparable relative loads, but the resistance represented by a bodyweight movement depends on body position and movement mechanics. [16] Where a defensible coefficient is unavailable, the model should use the user's own progression in that exact variation and avoid pretending that all bodyweight movements lift 100% of body mass.

Timed strength work needs a parallel route. For planks, wall sits and other isometrics, seconds under tension replace repetitions:

$$\text{Timed set dose} = \frac{\text{seconds}}{30} \times \text{relative intensity} \times \text{exercise involvement} \times \text{set difficulty}$$

Relative intensity can use external load where entered and a personal best hold for the same variation. Without either, body position and duration give a useful but lower-confidence estimate. Loaded carries sit between strength and locomotion: carried mass × distance is a useful external measure, exercise involvement supplies the muscular component, and wearable HRR independently supplies the cardiovascular component.

Cardio machines and outdoor endurance work are simpler because time and distance describe more of the task. Wearable HRR should remain the primary cardiovascular measure. Cycling power, where the device provides it, gives direct external work as average watts × seconds. Running and treadmill work can use distance, mean speed, gradient and elevation; rowing can use distance, pace or power. External work is particularly useful for filling short heart-rate gaps and checking whether a strangely low reading is plausible. It should estimate an intensity band rather than overwrite a good personal heart-rate response. Treadmill metabolic equations and generic MET (Metabolic Equivalent of Task is an estimate of how much energy different activities use compared to rest) values can support this fallback, but published equations show error across speeds, gradients and populations. [17] The updated Compendium of Physical Activities is also a useful category-level source when only the activity is known, while its values remain population estimates rather than a measurement of this user's session. [18]

A 45-minute "strength training" entry with no exercise detail should follow a different route. If heart rate is present, calculate cardio load normally across the session. Muscular load should begin with the user's median muscular points per active minute from their own recent detailed strength sessions, separated at least into lower body, upper body, full body and circuit training. Multiply that personal rate by the session's estimated active minutes, using set-like accelerometer bursts where the wearable provides them. Session RPE can adjust the result within a controlled range.

If the user has no detailed history, use a conservative category-and-duration prior. Traditional strength training should start below circuit training because total duration contains more rest, and a generic "strength" session should sit near the middle rather than assume a hard leg day. As detailed sessions accumulate, the population prior should be replaced by the user's own distribution. A duration-only estimate should retain a wider uncertainty range and should not become more precise just because it has been mapped to a neat integer. MET values may help estimate aerobic energy expenditure, although they cannot reconstruct which muscles were loaded or how close any set came to failure.

There are a few things the viable inputs still can't tell us. Range of motion changes mechanical work. Repetition tempo changes time under tension. Bar velocity loss is useful evidence of fatigue. Technique, instability, eccentric emphasis, blood-flow restriction and pain can all change the response to the same logged weight. More sensors or a short video could improve exercise recognition and movement quality, but none of that is really required for a useful first model. The important thing is to keep these unknowns in the confidence record rather than manufacture them from the exercise name.

Taken together, the calculation should work as follows:

- For running, cycling, rowing, treadmill work and cardio circuits: calculate cardiovascular load from the wearable HRR curve. Add external distance, speed, gradient, elevation or power as validation and as a fallback intensity estimate when HR is missing. These sessions receive little or no separate muscular load unless they include hills, sprints, resistance, carries or logged strength movements.

- For detailed dynamic strength training: calculate cardio load from HRR during the session. Calculate every set as repetitions × relative load × exercise-involvement coefficient × difficulty, allocate the result to the muscles involved, then apply diminishing returns when local and session totals are combined. Relative load comes from a slowly changing exercise-specific e1RM. Missing effort leaves difficulty at 1.0 and lowers confidence.

- For bodyweight strength training: replace entered weight with exercise-specific effective body mass, then use the same relative-load calculation. Keep each variation separate and prefer the user's own performance history when biomechanical coefficients are weak.

- For isometrics: use seconds under tension, external or effective bodyweight intensity, exercise involvement and set difficulty. For loaded carries: use carried mass × distance for muscular work and HRR for cardio work.

- For "strength training for 45 minutes": use HRR for the cardiovascular part. Estimate muscular load from active minutes × the user's recent muscular points per active minute for the closest session category. Until that personal history exists, use a conservative category prior with wide uncertainty and adjust it using session RPE (if given).

Cardiovascular and muscular load should be stored separately even if the interface later presents one total Strain number. This avoids us double counting and keeps the explanation honest: the session may be demanding because the heart worked for a long time, because a lot of local muscular work was completed, or because of both. Confidence should sit beside those loads and never change them. The coefficients above are reasonable starting values, not established biological constants. They should be tested against repeated user logs, post-session effort, next-day performance and recovery patterns before becoming fixed.

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